JESS Thermodynamic Database v8.9

Tungsten

24-Jan-23

Reaction No. 4899							
WO3(s) + H+1 + e-1 = 0.5<W2O5(s)> + 0.5<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.5(1SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.015(0.05SD)	0	MDG	775

Reaction No. 4900							
WO3(s) + 6<e-1> + 6<H+1> = W(s) + 3<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.09(0.01SD)	6	MDG	775
2	25	0	Inf. Dilution	E:-0.090(2SF)	5	CRV	6330
3	25	0	Inf. Dilution	E:-0.090(2SF)	5	CRV	22519
4	25	0	Inf. Dilution	E:-0.09(1SF)	4	CRV	22551

Reaction No. 4901							
WO4-2 + 6<e-1> + 4<H2O> = W(s) + 8<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-106.4(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.01(0.01SD)	0	MEF	775
3	25	0	Inf. Dilution	E:-1.074(4SF)	0	CRV	22551

Reaction No. 4902							
H+1(2)_WO4-2 + 6<H+1> + 6<e-1> = W(s) + 4<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.2(2SF)	1	ELC	

Reaction No. 4903							
0.5<W2O5(s)> + H+1 + e-1 = WO2(s) + 0.5<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.9(2SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.02(0.01SD)	0	MDG	775

Reaction No. 4904

doi: 10.5281/zenodo.7700024

WO2(s) + 4<H+1> + 4<e-1> = W(s) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.12(0.01SD)	0	MDG	775
3	25	0	Inf. Dilution	E:-0.119(3SF)	0	CRV	6330
4	25	0	Inf. Dilution	E:-0.119(3SF)	0	CRV	22519

Reaction No. 7668

2<WO4-2> + DLyxose + 2<H+1> = H+1(2)_DLyxose_WO4-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:18.1(0.06SD)	4	MGL	1070

Reaction No. 7669

2<WO4-2> + DMannose + 2<H+1> = H+1(2)_DMannose_WO4-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:17.5(0.06SD)	4	MGL	1070

Reaction No. 7672

2<WO4-2> + LRhamnose + 2<H+1> = H+1(2)_LRhamnose_WO4-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:17.0(0.06SD)	4	MGL	1070

Reaction No. 7673

2<WO4-2> + LRhamnose + 3<H+1> = H+1(3)_LRhamnose_WO4-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:20.5(0.1SD)	3	MGL	1070

Reaction No. 9379

MoO4-2 + WO3_NTA-3 = MoO3_NTA-3 + WO4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.0(2SF)	1	ELC	
2	28	1.3	Unknown	lgK:0.15(2SF)	6	MMR	1118

Reaction No. 9381

WO4-2 + NTA-3 + 2<H+1> = WO3_NTA-3 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.15	Unknown	lgK:18.9(0.05SD)	4	MGL	1118

2	25	0.5	NaClO4	lgK:17.8(0.03SD)	4	MGL	1118
3	25	3	NaClO4	lgK:19.0(0.2SD)	5	MPS	6969
4	35	1:2.5	Unknown	lgK:19.1(0.2SD)	3	MMR	1118

Reaction No. 17981

WO4-2 + IDA-2 + 2<H+1> = H+1(2)_IDA-2_WO4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.15	Unknown	lgK:18.5(0.2SD)	3	MGL	1397

Reaction No. 17982

WO4-2 + MIDA-2 + 2<H+1> = H+1(2)_MIDA-2_WO4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.15	Unknown	lgK:18.70(0.01SD)	6	MGL	1397

Reaction No. 17985

WO4-2 + NTA-3 + 2<H+1> = H+1(2)_WO4-2_NTA-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.15	Unknown	lgK:18.9(0.05SD)	4	MGL	1397

Reaction No. 17986

EDTA-4 + WO4-2 + 2<H+1> = H+1(2)_WO4-2_EDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.7(4SF)	1	EOR	
2	25	0.15	Unknown	lgK:19.(0.4SD)	3	MGL	1397

Reaction No. 17989

H+1(2)_WO4-2_EDTA-4 + WO4-2 + 2<H+1> = H+1(4)_WO4-2(2)_EDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.15	Unknown	lgK:16.9(0.2SD)	4	MGL	1397

Reaction No. 19280

IDA-2 + WO4-2 = IDA-2_WO4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaClO4	lgK:18.1(0.05SD)	4	MGL	1659

Reaction No. 19281

NTA-3 + WO4-2 = WO4-2_NTA-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	3	NaClO4	lgK:19.0(0.08SD)	3	MGL	1659

Reaction No. 19282							
EDDA-2 + WO4-2 = EDDA-2_WO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaClO4	lgK:19.6(0.06SD)	3	MGL	1659

Reaction No. 19283							
EDTA-4 + WO4-2 = WO4-2_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.38(4SF)	1	EOR	
2	25	3	NaClO4	lgK:19.7(0.05SD)	3	MGL	1659

Reaction No. 19284							
2<EDTA-4> + WO4-2 = WO4-2_EDTA-4(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:30.92(4SF)	1	EOR	
2	25	3	NaClO4	lgK:36.2(0.06SD)	4	MGL	1659

Reaction No. 29648							
Asp-2 + WO2+2 = WO2+2_Asp-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:8.20(3SF)	5	MGL	1814

Reaction No. 29649							
WO2+2_Asp-2 + Asp-2 = WO2+2_Asp-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:5.87(3SF)	5	MGL	1814

Reaction No. 29650							
WO2+2_Asp-2(2) + Asp-2 = WO2+2_Asp-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:3.81(3SF)	5	MGL	1814

Reaction No. 29651							
Glu-2 + WO2+2 = WO2+2_Glu-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:7.95(3SF)	5	MGL	1814

Reaction No. 29652

WO2+2_Glu-2 + Glu-2 = WO2+2_Glu-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:5.45(3SF)	5	MGL	1814

Reaction No. 29653

WO2+2_Glu-2(2) + Glu-2 = WO2+2_Glu-2(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:3.60(3SF)	5	MGL	1814

Reaction No. 30905

2<H+1> + 2<Lactic-1> + WO3 = H+1(2)_WO3_Lactic-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaCl	lgK:17.47(0.01SD)	5	MGL	1938
2	25	1	NaCl	dH:-80.(2.SD)kJ	5	MCL	1938

Reaction No. 30906

3<H+1> + 2<Lactic-1> + WO3 = H+1(3)_WO3_Lactic-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaCl	lgK:18.4(0.1SD)	5	MGL	1938

Reaction No. 30907

2<H+1> + Lactic-1 + WO3 = H+1(2)_WO3_Lactic-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaCl	lgK:13.0(0.07SD)	5	MGL	1938

Reaction No. 30908

3<H+1> + Lactic-1 + WO3 = H+1(3)_WO3_Lactic-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaCl	lgK:14.6(0.1SD)	5	MGL	1938

Reaction No. 30909

3<H+1> + 2<Lactic-1> + 2<WO3> = H+1(3)_WO3(2)_Lactic-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaCl	lgK:25.5(0.1SD)	5	MGL	1938

Reaction No. 36324

WO4-2 + Sulfox-2 + 2<H+1> = WO3_Sulfox-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15:17	0.1	KNO3	lgK:19.9(0.07SD)	4	MGL	906

Reaction No. 37542

WO4-2 + 2<H+1(2)_Cat-2> = WO2+2_Cat-2(2) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaHSO3	lgK:6.53(3SF)	5	MSP	140

Reaction No. 40407

H+1(2)_WO4-2 + H+1(2)_Oxalic-2 = H+1(2)_WO3_Oxalic-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.08:0.24	H2SO4	lgK:4.8(0.09SD)	4	MPL	931
2	25			lgK:5.1(0.05SD)	4	MKN	931

Reaction No. 40408

H+1(2)_WO3_Oxalic-2 + H+1(2)_Oxalic-2 = WO2+2_H+1(2)_Oxalic-2(2) + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.08:0.24	H2SO4	lgK:7.5(0.1SD)	4	MPL	931

Reaction No. 42161

WO4-2 + Cys-2 + 2<H+1> = WO3_Cys-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	1	NaCl	lgK:18.8(0.2SD)	5	MGL	2547

Reaction No. 43051

Malonic-2 + WO2+2 = WO2+2_Malonic-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:1.34(3SF)	3	ENS	906

Reaction No. 43217

Succinic-2 + WO2+2 = WO2+2_Succinic-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0.1	Unknown	lgK:1.06(3SF)	4	MIX	906

Reaction No. 44093

2<35DiNitrCat-2> + WO2+2 = WO2+2_35DiNitrCat-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0.1	NaNO3	lgK:20.74(4SF)	4	MSP	906

Reaction No. 44399

WO2+2 + 2<H+1_BenzHydroxam-1> = WO2+2_BenzHydroxam-1(2) + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	4	HCl	lgR:7.11(3SF)	4	MDS	906

Reaction No. 44896

WO4-2 + EDTA-4 + 2<H+1> = WO3_EDTA-4 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.15	Unknown	lgK:19.(0.4SD)	4	MGL	906
2	35	1:2.5	Unknown	lgK:19.(0.3SD)	4	MMR	906

Reaction No. 44897

WO4-2 + WO3_EDTA-4 + 2<H+1> = WO3(2)_EDTA-4 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.15	Unknown	lgK:16.9(0.2SD)	4	MGL	906
2	35	1:2.5	Unknown	lgK:17.(0.3SD)	4	MMR	906

Reaction No. 44898

WO3_EDTA-4 + H+1 = H+1_WO3_EDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.5(2SF)	1	ESC	
2	35	1:2.5	Unknown	lgK:7.5(0.2SD)	4	MMR	906

Reaction No. 54623

Pb(s) + W(s) + 2<O2(g)> = Pb+2_WO4-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:178.8(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:177.6(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:128.8(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:99.55(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:80.11(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-243.9(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-269.4(0.1SD)	4	MCL	5029
8	25	0	Inf. Dilution	dH:-268.1(4SF)	4	MTD	6422
9	127	0	Inf. Dilution	dH:-267.7(4SF)	4	MTD	6422

10	227	0	Inf. Dilution	dH:-267.2(4SF)	4	MTD	6422
11	327	0	Inf. Dilution	dH:-266.6(4SF)	4	MTD	6422

Reaction No. 55349

2<Ag(s)> + W(s) + 2<O2(g)> = Ag+1(2)_WO4-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-199.7(0.05SD)	4	MPT	5501
2	25	0	Inf. Dilution	dH:-231.5(0.1SD)	4	MPT	5501

Reaction No. 56379

Mg(s) + W(s) + 2<O2(g)> = Mg+2_WO4-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:246.0(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:244.4(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:178.4(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:138.9(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:112.5(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-339.6(0.05SD)	0	ENS	802
7	25	0	Inf. Dilution	dG:-335.6(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-366.3(0.1SD)	3	ENS	802
9	25	0	Inf. Dilution	dH:-362.3(4SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-362.1(4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-361.8(4SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-361.3(4SF)	4	MTD	6422

Reaction No. 56392

Sr(s) + W(s) + 2<O2(g)> = Sr+2_WO4-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:268.4(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:266.7(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:195.3(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:152.6(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:124.1(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-366.0(0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-366.0(0.05SD)	5	ENS	802
8	25	0	Inf. Dilution	dG:-366.2(4SF)	4	EFE	6422
9	25	0	Inf. Dilution	dH:-391.9(0.1SD)	4	ENS	802

10	25	0	Inf. Dilution	dH:-391.9(0.1SD)	5	ENS	802
11	25	0	Inf. Dilution	dH:-391.9(4SF)	4	MTD	6422
12	127	0	Inf. Dilution	dH:-391.5(4SF)	4	MTD	6422
13	227	0	Inf. Dilution	dH:-391.2(4SF)	4	MTD	6422
14	327	0	Inf. Dilution	dH:-390.8(4SF)	4	MTD	6422

Reaction No. 57516

WO4-2 + 2<H+1> = H+1(2)_WO4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:8.1(2SF)	0	CRV	552
2	25	0	Inf. Dilution	lgK:5.8(2SF)	0	CRV	552
3	25	3	Na(ClO4)	lgK:11.3(0.1SD)	4	MEF	5564

Reaction No. 57517

6<H+1> + 6<WO4-2> = W6O21-6 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:59.3(3SF)	1	EOR	
2	25	3	Na(ClO4)	lgK:52.5(0.1SD)	6	MEF	5564
3	25	3	NaClO4	dH:-57.(1.SD)	5	MCL	43

Reaction No. 57518

7<H+1> + 6<WO4-2> = H+1_W6O21-6 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:52.(2SF)	1	EOR	
2	25	3	Na(ClO4)	lgK:60.76(0.03SD)	6	MEF	5564
3	25	3	Na(ClO4)	lgK:60.68(0.03SD)	4	MGL	31135
4	25	3	NaClO4	dH:-62.(0.9SD)	5	MCL	43

Reaction No. 58982

Ag+1(2)_WO4-2_(s) + 2<e-1> = 2<Ag(s)> + WO4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.4660(4SF)	4	MAG	778
2	25	0	Inf. Dilution	E:0.466(3SF)	4	CRV	3006
3	25	0	Inf. Dilution	E:0.4660(4SF)	5	CRV	6330

Reaction No. 65644

H+1 + WO4-2 = H+1_WO4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.6(2SF)	4	CRV	552
2	25	0	Inf. Dilution	lgK:3.6(0.5MD)	4	MGL	7167
3	25	0.1	Unknown	lgK:3.(0.3SD)	3	CRV	552
4	25	0.5	Unknown	lgK:3.0(2SF)	0	CRV	552
5	25	1	NaCl	lgK:2.9(2SF)	3	CRV	552
6	25	3	NaCl	lgK:2.7(0.1SD)	3	CRV	552
7	150	0	Inf. Dilution	lgK:4.71(0.06MD)	3	MGL	7167
8	150	0.104	NaCl/m	lgK:4.09(0.06MD)	3	MGL	7167
9	150	1.014	NaCl/m	lgK:3.80(0.12MD)	3	MGL	7167
10	200	0.104	NaCl/m	lgK:4.68(0.03MD)	3	MGL	7167
11	200	1.014	NaCl/m	lgK:4.17(0.03MD)	3	MGL	7167
12	200	5.12	NaCl/m	lgK:4.06(0.12MD)	3	MGL	7167
13	250	0.104	NaCl/m	lgK:5.21(0.03MD)	3	MGL	7167
14	250	1.014	NaCl/m	lgK:4.58(0.03MD)	3	MGL	7167
15	250	5.12	NaCl/m	lgK:4.31(0.09MD)	3	MGL	7167
16	290	0.104	NaCl/m	lgK:5.64(0.03MD)	2	MGL	7167
17	290	1.014	NaCl/m	lgK:4.78(0.03MD)	2	MGL	7167
18	290	5.12	NaCl/m	lgK:4.46(0.09MD)	0	MGL	7167
19	20:300	0.1	Unknown	dH:1.(1SF)	0	CRV	552

Reaction No. 65645							
Pb+2_WO4-2_(s) = Pb+2 + WO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-14.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-16.07(4SF)	0	CRV	552
3	25	0	Inf. Dilution	lgK:-11.35(4SF)	0	CRV	28992
4	50	0	Inf. Dilution	lgK:-10.79(4SF)	0	CRV	28992
5	100	0	Inf. Dilution	lgK:-10.37(4SF)	0	CRV	28992
6	150	0	Inf. Dilution	lgK:-10.6(3SF)	0	CRV	28992
7	200	0	Inf. Dilution	lgK:-11.2(3SF)	0	CRV	28992
8	25	0	Inf. Dilution	dH:12.50(4SF)	1	CRV	552

Reaction No. 65646							
H+1_W6O21-6 = W6O21-6 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	3	NaClO4	lgK:8.30(3SF)	5	CRV	552
2	25	3	NaClO4	dH:-5.(1SF)	5	CRV	552

Reaction No. 65647							
2<Ag+1> + WO4-2 = Ag+1(2)_WO4-2_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0	Inf. Dilution	lgK:10.88(4SF)	2	MIS	6843
2	25	0	Inf. Dilution	lgK:12.12(4SF)	0	CRV	552
3	25	0	Inf. Dilution	lgK:12.32(4SF)	0	CRV	3006
4	25	0	Inf. Dilution	lgK:10.63(4SF)	2	MIS	6843
5	25	0.1	NaNO3	lgK:11.44(4SF)	6	CRV	552
6	25	0.5	NaNO3	lgK:11.08(4SF)	6	CRV	552
7	25	1	NaNO3	lgK:10.8(0.1SD)	6	CRV	552
8	25	0	Inf. Dilution	dH:14.7(3SF)	4	CRV	552

Reaction No. 66225							
Zn(s) + W(s) + 2<O2(g)> = Zn+2_WO4-2_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:196.8(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:195.4(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:142.0(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:110.0(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:88.66(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-268.4(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-293.7(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-293.2(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-292.7(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-292.1(4SF)	4	MTD	6422

Reaction No. 66271							
W(s) + 2<S(s)> = WS2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:43.78(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:40.81(4SF)	0	ENS	6494
3	27	0	Inf. Dilution	lgK:43.50(4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:40.55(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:32.17(4SF)	4	ENS	6422

6	127	0	Inf. Dilution	lgK:30.00(4SF)	0	ENS	6494
7	227	0	Inf. Dilution	lgK:25.23(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:23.53(4SF)	0	ENS	6494
9	327	0	Inf. Dilution	lgK:20.56(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:19.16(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-59.73(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-55.69(4SF)	0	EFE	6494
13	25	0	Inf. Dilution	dH:-62.0(3SF)	4	MTD	6422
14	25	0	Inf. Dilution	dH:-57.74(4SF)	0	MTD	6494
15	127	0	Inf. Dilution	dH:-63.2(3SF)	4	MTD	6422
16	127	0	Inf. Dilution	dH:-58.91(4SF)	0	MTD	6494
17	227	0	Inf. Dilution	dH:-63.9(3SF)	4	MTD	6422
18	227	0	Inf. Dilution	dH:-59.70(4SF)	0	MTD	6494
19	327	0	Inf. Dilution	dH:-64.5(3SF)	4	MTD	6422
20	327	0	Inf. Dilution	dH:-60.28(4SF)	0	MTD	6494

Reaction No. 66272

W(s) + 0.5<O2(g)> + 2<F2(g)> = WOF4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:225.2(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:223.7(4SF)	4	ENS	6422
3	106	0	Inf. Dilution	lgK:173.2(4SF)	4	ENS	6422
4	25	0	Inf. Dilution	dG:-307.2(4SF)	4	EFE	6422
5	25	0	Inf. Dilution	dH:-333.3(4SF)	4	MTD	6422
6	27	0	Inf. Dilution	dH:-333.2(4SF)	4	MTD	6422
7	106	0	Inf. Dilution	dH:-332.5(4SF)	4	MTD	6422

Reaction No. 66273

W(s) + O2(g) + Cl2(g) = WO2Cl2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:123.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:122.3(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:88.35(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:68.06(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:54.59(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-168.0(4SF)	3	EFE	6422

7	25	0	Inf. Dilution	dH:-186.5(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-186.0(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-185.3(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-184.4(4SF)	0	MTD	6422

Reaction No. 66274

W(s) + 0.5<O2(g)> + 2<Cl2(g)> = WOCl4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:96.23(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:95.50(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:66.38(4SF)	2	ENS	6422
4	211	0	Inf. Dilution	lgK:51.31(4SF)	1	ENS	6422
5	25	0	Inf. Dilution	dG:-131.3(4SF)	4	EFE	6422
6	25	0	Inf. Dilution	dH:-160.4(4SF)	2	MTD	6422
7	127	0	Inf. Dilution	dH:-159.4(4SF)	1	MTD	6422
8	211	0	Inf. Dilution	dH:-158.4(4SF)	0	MTD	6422

Reaction No. 66275

W(s) + 1.5<O2(g)> = WO3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:133.9(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:132.9(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:96.27(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:74.30(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:59.68(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-182.6(4SF)	3	EFE	6422
7	25	0	Inf. Dilution	dG:-182.6(4SF)	3	CRV	31291
8	25	0	Inf. Dilution	dH:-201.5(4SF)	3	MTD	6422
9	127	0	Inf. Dilution	dH:-201.2(4SF)	1	MTD	6422
10	227	0	Inf. Dilution	dH:-200.9(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-200.4(4SF)	0	MTD	6422

Reaction No. 66276

W(s) + 1.48<O2(g)> = WO2.96(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:132.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:131.7(4SF)	4	ENS	6422

3	127	0	Inf. Dilution	lgK:95.40(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:73.63(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:59.15(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-180.9(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-199.6(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-199.3(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-199.0(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-198.5(4SF)	4	MTD	6422

Reaction No. 66277

W(s) + 1.45<O2(g)> = WO2.90(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:130.3(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:129.4(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:93.69(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:72.32(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:58.10(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-177.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-196.0(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-195.8(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-195.4(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-195.0(4SF)	4	MTD	6422

Reaction No. 66278

W(s) + 1.36<O2(g)> = WO2.72(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:124.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:123.3(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:89.32(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:68.95(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:55.41(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-169.4(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-186.7(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-186.5(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-186.2(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-185.8(4SF)	4	MTD	6422

Reaction No. 66279							
O2(g) + W(s) = WO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:93.53(4SF)	3	ENS	6422
2	25	0	Inf. Dilution	lgK:93.53(4SF)	3	ENS	6494
3	27	0	Inf. Dilution	lgK:92.89(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:92.89(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:67.23(4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:67.24(4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:51.86(4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:51.86(4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:41.63(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:41.63(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-127.6(4SF)	3	EFE	6422
12	25	0	Inf. Dilution	dG:-127.6(4SF)	3	EFE	6494
13	25	0	Inf. Dilution	dG:-127.61(5SF)	3	CRV	31291
14	25	0	Inf. Dilution	dH:-140.9(4SF)	3	MTD	6422
15	25	0	Inf. Dilution	dH:-140.9(4SF)	3	MTD	6494
16	127	0	Inf. Dilution	dH:-140.8(4SF)	2	MTD	6422
17	127	0	Inf. Dilution	dH:-140.8(4SF)	1	MTD	6494
18	227	0	Inf. Dilution	dH:-140.6(4SF)	1	MTD	6422
19	227	0	Inf. Dilution	dH:-140.6(4SF)	2	MTD	6494
20	327	0	Inf. Dilution	dH:-140.3(4SF)	0	MTD	6422
21	327	0	Inf. Dilution	dH:-140.3(4SF)	0	MTD	6494

Reaction No. 66280							
W(s) + 3<Cl2(g)> = WCl6(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:79.80(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:79.16(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:53.43(4SF)	2	ENS	6422
4	177	0	Inf. Dilution	lgK:44.92(4SF)	1	ENS	6422
5	25	0	Inf. Dilution	dG:-108.9(4SF)	4	EFE	6422
6	25	0	Inf. Dilution	dH:-141.9(4SF)	2	MTD	6422
7	127	0	Inf. Dilution	dH:-140.6(4SF)	1	MTD	6422
8	177	0	Inf. Dilution	dH:-139.8(4SF)	0	MTD	6422

Reaction No. 66281

W(s) + 2.5<Cl2(g)> = WCl5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:70.39(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:69.84(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:47.61(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:34.41(4SF)	1	ENS	6422
5	253	0	Inf. Dilution	lgK:31.83(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-96.03(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-122.6(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-121.4(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-120.0(4SF)	0	MTD	6422
10	253	0	Inf. Dilution	dH:-119.6(4SF)	0	MTD	6422

Reaction No. 66282

W(s) + 2<Cl2(g)> = WCl4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:62.97(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:62.49(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:43.28(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:31.86(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:24.33(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-85.91(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-105.9(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-105.0(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-104.0(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-102.9(4SF)	0	MTD	6422

Reaction No. 66283

W(s) + Cl2(g) = WCl2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.54(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:38.26(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:27.10(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:20.46(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:16.06(4SF)	0	ENS	6422

6	25	0	Inf. Dilution	dG:-52.58(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-61.5(3SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-61.0(3SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-60.5(3SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-60.0(3SF)	0	MTD	6422

Reaction No. 66284

2<W(s)> + C(s) = W2C(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.835(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:3.806(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:2.702(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:2.100(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:1.740(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-5.23(3SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-6.3(2SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-5.8(2SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-5.2(2SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-4.6(2SF)	4	MTD	6422

Reaction No. 66285

W(s) + C(s) = WC(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.723(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:6.679(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:4.937(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:3.902(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:3.219(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-9.17(3SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-9.6(2SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-9.5(2SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-9.4(2SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-9.3(2SF)	4	MTD	6422

Reaction No. 66286

W(s) + 6<C(s)> + 3<O2(g)> = W(CO)6(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:148.5(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:147.4(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:106.2(4SF)	4	ENS	6422
4	25	0	Inf. Dilution	dG:-202.5(4SF)	4	EFE	6422
5	25	0	Inf. Dilution	dH:-227.5(4SF)	4	MTD	6422
6	127	0	Inf. Dilution	dH:-225.6(4SF)	4	MTD	6422

Reaction No. 66287

W(s) + 3<Br2(l)> = WBr6(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:50.94(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:50.57(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:33.25(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:22.08(4SF)	1	ENS	6422
5	309	0	Inf. Dilution	lgK:15.88(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-69.49(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-82.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-102.9(4SF)	0	MTD	6422
9	227	0	Inf. Dilution	dH:-101.4(4SF)	0	MTD	6422
10	309	0	Inf. Dilution	dH:-100.0(4SF)	0	MTD	6422

Reaction No. 66288

W(s) + 2.5<Br2(l)> = WBr5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:47.23(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:46.90(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:31.34(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:21.36(4SF)	1	ENS	6422
5	286	0	Inf. Dilution	lgK:17.20(4SF)	1	ENS	6422
6	25	0	Inf. Dilution	dG:-64.44(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-74.5(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-91.9(3SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-90.6(3SF)	0	MTD	6422
10	286	0	Inf. Dilution	dH:-89.8(3SF)	0	MTD	6422

Reaction No. 66519

Ba(s) + W(s) + 2<O2(g)> = Ba+2_WO4-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:280.0(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:278.2(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:204.1(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:159.7(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:130.1(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-382.1(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-407.0(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-406.5(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-406.1(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-406.0(4SF)	4	MTD	6422

Reaction No. 66533

Be(s) + W(s) + 2<O2(g)> = Be+2_WO4-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:246.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:244.5(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:178.6(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:139.1(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:112.8(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-335.8(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-361.7(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-361.6(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-361.3(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-360.9(4SF)	4	MTD	6422

Reaction No. 66599

Ca(s) + W(s) + 2<O2(g)> = Ca+2_WO4-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:266.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:269.5(4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:267.7(4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:196.2(4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:153.3(4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:124.7(4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-367.7(4SF)	0	EFE	6422

8	25	0	Inf. Dilution	dH:-393.2(4SF)	0	MTD	6422
9	127	0	Inf. Dilution	dH:-392.8(4SF)	0	MTD	6422
10	227	0	Inf. Dilution	dH:-392.4(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-391.9(4SF)	0	MTD	6422

Reaction No. 66600

3<Ca(s)> + W(s) + 3<O2(g)> = 3CaO.WO3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:493.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:490.4(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:360.4(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:282.4(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:230.5(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-673.4(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-714.1(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-713.6(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-712.8(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-712.0(4SF)	4	MTD	6422

Reaction No. 66617

Cd(s) + W(s) + 2<O2(g)> = Cd+2_WO4-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:189.0(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:187.7(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:136.4(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:105.6(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:85.14(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-257.9(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-282.1(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-281.7(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-281.3(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-282.4(4SF)	4	MTD	6422

Reaction No. 66650

Co(s) + W(s) + 2<O2(g)> = Co+2_WO4-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:181.5(4SF)	4	ENS	6422

2	27	0	Inf. Dilution	lgK:180.2(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:130.8(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:101.2(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:81.50(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-247.6(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-271.7(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-271.2(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-270.6(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-270.0(4SF)	4	MTD	6422

Reaction No. 66736

Fe(s) + W(s) + 2<O2(g)> = Fe+2_WO4-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:189.8(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:188.6(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:137.0(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:106.1(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:85.51(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-259.0(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-283.1(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-282.9(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-282.8(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-282.6(4SF)	4	MTD	6422

Reaction No. 66844

H+1(2)_WO4-2_(s) = WO4-2 + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-15.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-14.05(4SF)	0	ENS	773

Reaction No. 66879

Hg2+2_WO4-2_(s) = Hg2+2 + WO4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-16.96(4SF)	4	ENS	773

Reaction No. 67007

Mn(s) + 2<O2(g)> + W(s) = Mn+2_WO4-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:208.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:211.2(4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:209.8(4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:153.0(4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:119.0(4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:96.31(4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-288.1(4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dH:-311.9(4SF)	0	MTD	6422
9	127	0	Inf. Dilution	dH:-311.5(4SF)	0	MTD	6422
10	227	0	Inf. Dilution	dH:-311.1(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-310.7(4SF)	0	MTD	6422

Reaction No. 67062

2<Na(s)> + 2<O2(g)> + W(s) = Na+1(2)_WO4-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:250.5(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:248.8(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:181.5(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:141.1(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:114.1(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-341.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-369.2(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-370.4(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-370.0(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-369.5(4SF)	4	MTD	6422

Reaction No. 67121

Ni(s) + 2<O2(g)> + W(s) = Ni+2_WO4-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:179.3(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:178.1(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:129.1(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:99.68(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:80.10(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-244.7(4SF)	4	EFE	6422

7	25	0	Inf. Dilution	dH:-269.4(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-269.1(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-268.9(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-268.7(4SF)	4	MTD	6422

Reaction No. 67311

H2(g) + 2<O2(g)> + W(s) = H+1(2)_WO4-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:175.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:174.7(4SF)	4	ENS	6422
3	120	0	Inf. Dilution	lgK:128.0(4SF)	4	ENS	6422
4	25	0	Inf. Dilution	dG:-239.9(4SF)	4	EFE	6422
5	25	0	Inf. Dilution	dH:-270.5(4SF)	4	MTD	6422
6	120	0	Inf. Dilution	dH:-270.3(4SF)	4	MTD	6422

Reaction No. 70583

WO4-2 + 2<H2O> = OH-1(2)_WO4-2 + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	200	5.12	NaCl/m	lgK:7.64(0.12MD)	4	MGL	7167
2	250	5.12	NaCl/m	lgK:7.42(0.12MD)	4	MGL	7167
3	290	5.12	NaCl/m	lgK:7.27(0.11MD)	4	MGL	7167

Reaction No. 70584

6<WO4-2> + 7<H2O> = OH-1(7)_WO4-2(6) + 7<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	95	1.014	NaCl/m	lgK:48.14(0.06MD)	6	MGL	7167
2	150	1.014	NaCl/m	lgK:44.33(0.18MD)	0	MGL	7167
3	200	1.014	NaCl/m	lgK:42.78(0.18MD)	0	MGL	7167
4	250	1.014	NaCl/m	lgK:43.09(0.21MD)	0	MGL	7167
5	290	1.014	NaCl/m	lgK:44.09(0.24MD)	0	MGL	7167

Reaction No. 70585

6<WO4-2> + 10<H2O> = OH-1(10)_WO4-2(6) + 10<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	150	1.014	NaCl/m	lgK:58.05(0.18MD)	6	MGL	7167
2	200	0.104	NaCl/m	lgK:59.31(0.09MD)	0	MGL	7167
3	200	1.014	NaCl/m	lgK:55.90(0.18MD)	0	MGL	7167

4	250	0.104	NaCl/m	lgK:60.74(0.33MD)	0	MGL	7167
5	250	1.014	NaCl/m	lgK:55.55(0.24MD)	0	MGL	7167
6	290	0.104	NaCl/m	lgK:62.69(0.24MD)	0	MGL	7167
7	290	1.014	NaCl/m	lgK:55.81(0.30MD)	0	MGL	7167

Reaction No. 70586

12<WO4-2> + 18<H2O> = OH-1(18)_WO4-2(12) + 18<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	95	1.014	NaCl/m	lgK:117.33(0.36MD)	4	MGL	7167
2	150	0.104	NaCl/m	lgK:115.53(0.18MD)	0	MGL	7167
Note (2): Does not fit UCC model (and cannot apparently)							
3	150	1.014	NaCl/m	lgK:111.23(0.18MD)	0	MGL	7167
4	200	1.014	NaCl/m	lgK:107.20(0.15MD)	3	MGL	7167
5	200	5.12	NaCl/m	lgK:107.93(1.50MD)	0	MGL	7167

Reaction No. 70923

2<WO3(s)> + 2<H+1> + 2<e-1> = W2O5(s) + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.029(2SF)	5	CRV	6330
2	25	0	Inf. Dilution	E:-0.029(2SF)	5	CRV	22519

Reaction No. 70924

WO3(s) + 2<H+1> + 2<e-1> = WO2(s) + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.036(2SF)	5	CRV	6330
2	25	0	Inf. Dilution	E:0.036(2SF)	5	CRV	22519

Reaction No. 70925

W2O5(s) + 2<H+1> + 2<e-1> = 2<WO2(s)> + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.7(2SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.031(2SF)	0	CRV	6330
3	25	0	Inf. Dilution	E:-0.031(2SF)	0	CRV	22519

Reaction No. 70926

W+3 + 3<e-1> = W(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	E:0.1(1SF)	5	CRV	6330
2	25	0	Inf. Dilution	E:0.1(1SF)	3	CRV	7761
3	25	0	Inf. Dilution	E:0.1(1SF)	5	CRV	22519

Reaction No. 71288

Cu(s) + 2<O2(g)> + W(s) = Cu+2_WO4-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-1105.0(5SF)kJ	5	CRV	6330

Reaction No. 72379

2<O2(g)> + W(s) + 2<e-1> = WO4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-219.15(5SF)	3	CRV	6488
2	25	0	Inf. Dilution	dG:-916.9(4SF)kJ	5	CRV	7421
3	25	0	Inf. Dilution	dG:-218.500(6SF)	1	CRV	9029
4	25	0	Inf. Dilution	dG:-219.15(5SF)	3	CRV	10174
5	50	0	Inf. Dilution	dG:-219.35(5SF)	3	CRV	10174
6	75	0	Inf. Dilution	dG:-219.48(5SF)	3	CRV	10174
7	100	0	Inf. Dilution	dG:-219.55(5SF)	3	CRV	10174
8	125	0	Inf. Dilution	dG:-219.54(5SF)	3	CRV	10174
9	150	0	Inf. Dilution	dG:-219.46(5SF)	3	CRV	10174
10	175	0	Inf. Dilution	dG:-219.29(5SF)	3	CRV	10174
11	200	0	Inf. Dilution	dG:-219.01(5SF)	3	CRV	10174
12	225	0	Inf. Dilution	dG:-218.60(5SF)	3	CRV	10174
13	250	0	Inf. Dilution	dG:-218.01(5SF)	3	CRV	10174
14	300	0	Inf. Dilution	dG:-216.03(5SF)	0	CRV	10174
15	25	0	Inf. Dilution	dH:-257.1(4SF)	0	CRV	6488
16	25	0	Inf. Dilution	dH:-236.500(6SF)	0	CRV	9029

Reaction No. 72380

0.5<H2(g)> + 2<O2(g)> + W(s) + e-1 = H+1_WO4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:164.1(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-223.400(6SF)	0	CRV	9029
3	25	0	Inf. Dilution	dH:-255.000(6SF)	2	CRV	9029

Reaction No. 72718

2<H+1> + 2<Mandelic-1> + WO3 = H+1(2)_WO3_Mandelic-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaCl	lgK:17.59(0.01MD)	5	MGL	5810

Reaction No. 72719

3<H+1> + 2<Mandelic-1> + WO3 = H+1(3)_WO3_Mandelic-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaCl	lgK:18.09(0.30MD)	5	MGL	5810

Reaction No. 72720

2<H+1> + Mandelic-1 + WO3 = H+1(2)_WO3_Mandelic-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaCl	lgK:13.78(0.07MD)	5	MGL	5810

Reaction No. 72721

3<H+1> + Mandelic-1 + WO3 = H+1(3)_WO3_Mandelic-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaCl	lgK:15.33(0.07MD)	5	MGL	5810

Reaction No. 72811

WO4-2 + 2<Glycolic-1> + 2<H+1> = WO2+2_H+1(-2)_Glycolic-1(2) + 2<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaCl	lgK:16.85(0.14SD)	4	MMR	5801

Reaction No. 72827

WO4-2 + 2<Lactic-1> + 2<H+1> = WO2+2_H+1(-2)_Lactic-1(2) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaCl	lgK:18.15(0.14SD)	4	MMR	5801

Reaction No. 72828

WO4-2 + 2<2OHButan-1> + 2<H+1> = WO2+2_H+1(-2)_2OHButan-1(2) + 2<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaCl	lgK:18.25(0.14SD)	4	MMR	5801

Reaction No. 72829

WO4-2 + 2<Mandelic-1> + 2<H+1> = WO2+2_H+1(-2)_Mandelic-1(2) + 2<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaCl	lgK:18.15(0.14SD)	4	MMR	5801

Reaction No. 72830

WO4-2 + 2<2OH2MePropan-1> + 2<H+1> = WO2+2_H+1(-2)_2OH2MePropan-1(2) + 2<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaCl	lgK:18.75(0.14SD)	4	MMR	5801

Reaction No. 74271

WO2+2_Cl-1(3) + e-1 + 2<H+1> + 2<Cl-1> = WO+3_Cl-1(5) + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	12	HCl	E:0.36(2SF)	0	CRV	22551

Reaction No. 74272

2<WO+3_Cl-1(5)> + 4<e-1> + 4<H+1> = W2Cl9-3 + 2<H2O> + Cl-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	12	HCl	E:-0.05(1SF)	0	CRV	22551

Note (1): Reaction unbalanced in source (has only 1<H2O>)

Reaction No. 74273

W+5_CN-1(8) + e-1 = W+4_CN-1(8)

Note: Species W+4 and W+5 are unusual

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	E:0.457(3SF)	0	CRV	22551

Note (1): Conditional constant given without conditions

Reaction No. 74274

W+5_CN-1(4)_OH-1(4) + e-1 = W+4_CN-1(4)_OH-1(4)

Note: Species W+4 and W+5 are unusual

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	E:-0.74(2SF)	0	CRV	22551

Note (1): pH = 3.7; otherwise conditional constant without conditions

Reaction No. 78227

EDTA-4 + 2<WO4-2> + 4<H+1> = H+1(4)_WO4-2(2)_EDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:37.3(4SF)	1	ELC	

Reaction No. 78907

Zn+2 + WO4-2 = Zn+2_WO4-2_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.606(4SF)	1	ELC	

Reaction No. 79582							
Ca+2 + WO4-2 = Ca+2_WO4-2_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.80(3SF)	4	CRV	28992
2	50	0	Inf. Dilution	lgK:8.55(3SF)	3	CRV	28992
3	100	0	Inf. Dilution	lgK:8.59(3SF)	3	CRV	28992
4	150	0	Inf. Dilution	lgK:9.19(3SF)	2	CRV	28992
5	200	0	Inf. Dilution	lgK:10.1(3SF)	2	CRV	28992

Reaction No. 79583							
Fe+2 + WO4-2 = Fe+2_WO4-2_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:11.11(4SF)	0	CRV	28992
3	50	0	Inf. Dilution	lgK:10.86(4SF)	0	CRV	28992
4	100	0	Inf. Dilution	lgK:10.89(4SF)	0	CRV	28992
5	150	0	Inf. Dilution	lgK:11.35(4SF)	0	CRV	28992
6	200	0	Inf. Dilution	lgK:12.1(3SF)	0	CRV	28992

Reaction No. 79584							
Mn+2 + WO4-2 = Mn+2_WO4-2_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.50(3SF)	4	CRV	28992
2	50	0	Inf. Dilution	lgK:7.41(3SF)	3	CRV	28992
3	100	0	Inf. Dilution	lgK:7.68(3SF)	2	CRV	28992
4	150	0	Inf. Dilution	lgK:8.34(3SF)	2	CRV	28992
5	200	0	Inf. Dilution	lgK:9.27(3SF)	2	CRV	28992

Data Assessment

Weight	Assessment

9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
EFE	Free Energy relations
ELC	Linear Combination of Reactions
ENS	Not specified
EOR	Other Reaction Constants
ESC	Standard conditions anchor
MAG	Silver electrode e.m.f.
MCL	Calorimetry
MDG	Combination of Thermodynamic Data
MDS	Distribution between two phases
MEF	Electromotoric force, not specified
MGL	Glass electrode
MIS	Ion selective electrode
MIX	Ion exchange
MKN	Rate of reaction
MMR	Nuclear magnetic resonance
MPL	Polarography
MPS	Potentiometry and Spectrophotometry
MPT	Potentiometry
MSP	Spectroscopy

MTD	Temperature difference

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